

Actual thresholds and a direct coupled two-layer viscous return

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1 Reaching both original growth targets with retained diffusion

We now join the exact evolving-parent growth calculation to the direct second return theorem. Both stopping events are the original temperature thresholds. This changes the first growth time from the earlier fixed-inviscid-time variant in Section 3. The construction below specifies the frequencies and a nonempty interval of positive physical diffusivities, and proves that their actual entry states satisfy every inequality in Theorem 2. The spatial fields and their full forces are supplied by the compact construction following that theorem. In particular the amplitude damping on the released linear profile has its explicitly stated core force.

1.1 A finite choice within the original frequency schedule

Retain $A_0 \geq 1$, $\nu = \sqrt{A_0}$, $\mu = \lambda_0 = \lceil \sqrt{A_0}/2 \rceil$, and all fixed source profiles, including $s_*, s_0, C_\ell, \eta, \eta_2, \Lambda_1$. Thus $0 < s_* \leq 1/8$, $C_\ell \geq 4$, $H = \|\eta_2\|_\infty \geq 1$, $\Lambda_1 \geq 6H$, and $3\Lambda_1 H \sin s_* \leq 1/4$. The first section's λ_1 denoted a source frequency; in this section we write that same quantity as $\hat{\lambda}_1$ and retain the physical frequencies separately as $K_j = \mu \hat{\lambda}_j$. Keep the source's $Q_* \geq 200$, $Q_j = Q_* + j$, its specified finite constants $K(k)$, and the exact choices

$$\begin{aligned} \hat{\lambda}_2 &= \hat{\lambda}_1^{Q_2}, & k_j &= \max\{k \in \mathbb{N}_0 : 120k \leq Q_j, K(k) \leq \log \hat{\lambda}_j\}, \\ L_j &= (k_j + 41/8) \log \hat{\lambda}_j, & S_j &= \frac{A_0}{\mu} \hat{\lambda}_j^{-k_j-6}, \\ P_j^* &= \frac{A_0}{\mu} \hat{\lambda}_j^{-7/8} = S_j e^{L_j}. \end{aligned} \tag{1}$$

Here $K(k)$ is the source's prescribed parameter sequence, not the physical frequency K_j . The maximum is a nonempty finite set once $\log \hat{\lambda}_1 \geq K(0)$: $k = 0$ belongs to it and $k \leq \lfloor Q_j/120 \rfloor$. No smooth dependence of k_j on the frequencies is required.

The following fixed constants give one explicit sufficient frequency choice; their large numerical size is immaterial to finite existence:

$$\begin{aligned} C_A &= e^{2+\Lambda_1^{-1}}, & C_\sigma &= \sqrt{1+C_A}, & C_N &= 1 + 3C_\sigma/32, \\ C_L &= (\lfloor Q_2/120 \rfloor + 41/8)Q_2, \\ Y_0 &= 1 + \max \left\{ \begin{array}{l} 64, K(0), Q_2, 72 \cdot 16^3 H C_L^2 / \sqrt{e}, \quad 8 \log(4/e), \\ (16/47) \log(4C_\ell C_N/s_0), \quad (16/31) \log(2C_N/s_0) \end{array} \right\}. \end{aligned} \tag{2}$$

Choose any $\hat{\lambda}_1 \geq e^{Y_0}$ and then use (1) literally. Write $L = L_2$, $\Gamma_1 = \nu \sin s_*$ and $C_{\text{post}} = 2 + \Lambda_1^{-1}$. Define the explicit positive number

$$d_{\max} = \min \left\{ \frac{\Gamma_1}{2K_1^2}, \frac{\Gamma_1 \log 2}{K_1^2 C_{\text{post}}}, \frac{\nu \log(3/2)}{16K_1^2}, \frac{L\nu \log 3}{12800K_2^2} \right\}. \tag{3}$$

Every entry in this minimum uses only the fixed original data and the chosen frequencies. In particular it does not require solving a circular inequality involving a yet unknown diffusive

growth scale. All subsequent statements hold for each $0 < d \leq d_{\max}$, with physical viscosity and thermal diffusivity both equal to d .

1.2 First actual threshold and the changed first endpoint

Put $q = dK_1^2$ and $t_1^{\text{in}} = 1/\nu$. Growth and transition still have $\alpha = 0$. The signed first pair is exactly

$$\Theta_1(t) = -S_1 e^{(\Gamma_1 - q)(t - t_1^{\text{in}})}, \quad \Omega_1(t) = \frac{K_1}{\nu} \Theta_1(t). \quad (4)$$

Indeed its temperature coupling is $a_1 = A_0 \sin s_*/K_1$, its vorticity coupling is $b_1 = K_1 \sin s_*$, and $a_1 K_1/\nu = \Gamma_1$, $b_1 \nu/K_1 = \Gamma_1$. Differentiation verifies both damped equations and both original seed values. The linear integral equation proves uniqueness. Since $q \leq \Gamma_1/2$, the original first threshold is attained uniquely and transversely at

$$t_1^a = t_1^{\text{in}} + \frac{L_1}{\Gamma_1 - q}, \quad t_1^1 = t_1^a + \Gamma_1^{-1}, \quad t_1^b = t_1^1 + \Gamma_1^{-1}(1 + \Lambda_1^{-1}). \quad (5)$$

The first activation $\eta(\Gamma_1(t - t_1^{\text{in}}))$ is already one at t_1^a , since $L_1 \geq 1$.

Starting at t_1^a , take the original first transition and steering control with their displayed durations, translated to these actual times. The inverse map

$$(\Theta_1, \Omega_1)(t) = e^{-q(t - t_1^a)}(\Theta_{1,E}, \Omega_{1,E})(t)$$

relates their damped solution to the inviscid controlled solution with data

$$(\Theta_{1,E}, \Omega_{1,E})(t_1^a) = (-P_1^*, -K_1 P_1^*/\nu).$$

Differentiation gives the two damped equations, and multiplication by $e^{q(t - t_1^a)}$ gives the inverse equations and identical data. Both factors are positive. Thus the quotient, its entire trial family, and its unique first return root proved in Section 2 are unchanged on these translated intervals. At the selected endpoint

$$\alpha = s_*, \quad \zeta_1 = e_2, \quad \Omega_1 = 0, \quad A_d := -K_1 \Theta_1(t_1^b) = A_E e^{-qC_{\text{post}}/\Gamma_1}, \quad (6)$$

where the proved inviscid endpoint satisfies

$$2eA_0\hat{\lambda}_1^{1/8} \leq A_E \leq C_A A_0\hat{\lambda}_1^{1/8}. \quad (7)$$

This expression does not damp the inviscid growth duration: that duration has been replaced by the actual threshold time in (5). The first three diffusion exponents are exactly $qL_1/(\Gamma_1 - q)$ in growth, q/Γ_1 in transition, and $q(1 + \Lambda_1^{-1})/\Gamma_1$ in steering. The physical gains are respectively L_1 , $1 - q/\Gamma_1$, and the original selected steering gain minus the last exponent.

1.3 Actual second scale and all fixed design inequalities

The true inherited gradient at $t_2^{\text{in}} = t_1^b$ is $G_d = (A_0 \sin s_*, -A_0 \cos s_* - A_d)$. Evaluate the source scale at this changed endpoint, as required by its original rule:

$$\sigma^2 = |G_d| = \sqrt{A_0^2 + A_d^2 + 2A_0 A_d \cos s_*}, \quad s = s_2 = \frac{L\nu}{\sigma}, \quad a = \sin s, \quad \Gamma = \Gamma_2 = \sigma a. \quad (8)$$

Since $qC_{\text{post}}/\Gamma_1 \leq \log 2$, equations (7)–(8) imply

$$eA_0\hat{\lambda}_1^{1/8} \leq A_d \leq C_A A_0\hat{\lambda}_1^{1/8}, \quad \sqrt{e}\nu\hat{\lambda}_1^{1/16} \leq \sigma \leq C_\sigma \nu\hat{\lambda}_1^{1/16}. \quad (9)$$

For the lower scale bound use $\sigma^2 \geq A_d$, since the square of its left side differs from A_d^2 by a positive quantity. For the upper bound use the triangle inequality $\sigma^2 \leq A_0 + A_d \leq (1 + C_A)A_0\hat{\lambda}_1^{1/8}$. The choice $Y_0 > 8 \log(4/e)$ also gives $A_d > 4A_0$.

Let $y = \log \hat{\lambda}_1 \geq Y_0$. Then $64 < L \leq C_L y$. The exponential series with nonnegative terms proves $e^{y/16} \geq y^3/(6 \cdot 16^3)$; by (2) its right side is strictly greater than $12HC_L^2 y^2/\sqrt{e}$. Consequently

$$0 < s \leq \frac{L}{\sqrt{e} e^{y/16}} < \frac{1}{12HL} \leq \frac{1}{8}, \quad 0 < a \leq \frac{1}{8}, \quad 3LHa < \frac{1}{4}. \quad (10)$$

For $0 \leq s \leq 1/8$, integrating $\cos r \geq 1 - r^2/2 \geq 1/2$ on $[0, s]$ gives $s/2 \leq \sin s \leq s$. Thus

$$\frac{L\nu}{2} \leq \Gamma \leq L\nu. \quad (11)$$

The fixed base coefficients in the second-return coordinates obey

$$0 < C_0 = \frac{A_0 \cos s_*}{\sigma^2} < \frac{1}{4}, \quad 0 < C_1 = \frac{A_0 \sin s_*}{\sigma^2 a} \leq \frac{2\nu \sin s_*}{\sigma L} < \frac{1}{32}. \quad (12)$$

The last inequality uses $\sigma \geq \nu$, $L \geq 64$ and $\sin s_* < 1$; it has no missing comparison constant.

The same frequency choice proves the source support nesting for the entire second return and its finite hold. Set $N = 1 + (3 - \cos s)/a$. The exact matrix reconstruction in Theorem 2 gives both $\|M_2\|$ and $\|M_2^{-1}\|$ at most N , by the triangle inequality for its displayed shear. Equations (9) and $a \geq s/2$ imply $1/a \leq 2C_\sigma \hat{\lambda}_1^{1/16}/L$, whence $N \leq C_N \hat{\lambda}_1^{1/16}$. The final two logarithmic bounds in (2) give, strictly,

$$N\hat{\lambda}_1^{-3} < \frac{s_0}{2C_\ell}, \quad N\hat{\lambda}_1^{-2} < s_0. \quad (13)$$

Indeed their respective left sides are at most $C_N e^{-47y/16}$ and $C_N e^{-31y/16}$, and our choices bound these by $s_0/(4C_\ell)$ and $s_0/2$. For $\ell_1 = s_0/(C_\ell \mu)$ and $\ell_2 = \hat{\lambda}_1^{-3}/\mu$, this places the entire second support inside the first envelope's constant region and its exact linear profile region: $|x| \leq N\ell_2 < \ell_1/2$ and $|K_1 \zeta_1 \cdot x| \leq K_1 N\ell_2 < s_0$. Every original factor of μ is present in these identities.

1.4 Second growth with the actual diffusing parent

Write $r = t - t_2^{\text{in}}$. The second growth and transition have $\alpha = s_*$, $\Omega_1 = 0$, $M_2 = I$, $\zeta_2 = e(s)$ and $A_1(r) = A_d e^{-qr}$. These claims follow from the closed subsystem: $\zeta_{1,1} = 0$ makes $\dot{\Omega}_1 = -q\Omega_1 = 0$; both controls F_1^{ctl} and F_2^{ctl} are then zero, and the phase matrix equation is $\dot{M}_2 = 0$. The first temperature equation is $\dot{A}_1 = -qA_1$. The second pair has the exact coefficients

$$a_2(r) = \frac{A_d e^{-qr} \sin s + A_0 \sin(s_* + s)}{K_2}, \quad b_2 = K_2 \sin s, \quad d_2^{\text{phys}} = dK_2^2, \quad (14)$$

and the exact original entry data $\Theta_2(0) = -S_2$, $\Omega_2(0) = \sqrt{b_2/a_2(0)}\Theta_2(0)$. This is precisely the system solved by the convergent fundamental functions in Section 3; the substitution here uses its displayed changed coefficients, not frozen values.

Put $a_i = a_2(0)$, $u_i = \sqrt{b_2/a_i}$, $\gamma_i = \sqrt{a_i b_2}$ and

$$n_i = \frac{\gamma_i^2}{\Gamma^2} = \frac{A_d + A_0 \cos s_*}{\sigma^2} + C_1 \cos s.$$

One has

$$\frac{4}{5} < n_i < \frac{33}{32} < 2. \quad (15)$$

The lower bound follows from $A_d/\sigma^2 \geq A_d/(A_d + A_0) > 4/5$. For the upper bound, square $\sigma^2 = \sqrt{(A_d + A_0 \cos s_*)^2 + A_0^2 \sin^2 s_*}$ to get $(A_d + A_0 \cos s_*)/\sigma^2 \leq 1$, then use (12). In particular $\gamma_i \geq \Gamma/\sqrt{2} \geq L\nu/(2\sqrt{2})$.

On the explicitly fixed horizon $0 \leq r \leq 16/\nu$, the third bound in (3) gives $qr \leq \log(3/2)$. Since both terms in a_2 are positive, it follows that

$$\frac{1}{2}a_i \leq a_2(r) \leq a_i. \quad (16)$$

To prove all signs and ratio estimates without assuming them, set $X = -e^{d_2^{\text{phys}} r} \Theta_2$ and $Y = -e^{d_2^{\text{phys}} r} \Omega_2$. They solve $X' = a_2 Y$, $Y' = b_2 X$ with positive initial values. Before any first zero, both derivatives are positive, excluding that zero by continuity. The linear integral equation, whose coefficient matrix is bounded on this horizon, gives existence there by its uniformly convergent iterated-integral series. Therefore $u = Y/X = \Omega_2/\Theta_2$ exists and satisfies $u' = b_2 - a_2 u^2$. The constant functions u_i and $\sqrt{2}u_i$ are respectively subsolution and supersolution on this horizon, because $b_2 - a_2 u_i^2 \geq 0$ and $b_2 - 2a_2 u_i^2 \leq 0$. Subtracting either comparison equation from the equation for u gives the linear difference coefficient $-a_2(u + u_{\text{comparison}})$; its integrating factor preserves the stated ordering. Hence

$$u_i \leq u \leq \sqrt{2}u_i, \quad \frac{1}{\sqrt{n_i}} \leq v := \frac{\sigma\Omega_2}{K_2\Theta_2} \leq \sqrt{\frac{2}{n_i}}, \quad \frac{1}{2} < v < 2. \quad (17)$$

Here $\sigma u_i/K_2 = \Gamma/\gamma_i = 1/\sqrt{n_i}$; the equality follows by substituting the original a_i, b_2, K_2 .

For $P_2 = -\Theta_2$, the exact physical logarithmic derivative is

$$(\log P_2)' = a_2 u - d_2^{\text{phys}}.$$

The last bound in (3) gives $d_2^{\text{phys}} \leq L\nu \log 3/12800 < L\nu/16 \leq \gamma_i/4$. Thus throughout the fixed horizon

$$\frac{\gamma_i}{4} \leq (\log P_2)' \leq \sqrt{2}\gamma_i \leq 2\Gamma. \quad (18)$$

Indeed $a_2 u \geq a_i u_i/2 = \gamma_i/2$ and $a_2 u \leq \sqrt{2}a_i u_i = \sqrt{2}\gamma_i$. Consequently the original second target $P_2^* = S_2 e^L$ is reached at exactly one $r = r_a$, with nonzero derivative, and

$$\frac{L}{2\Gamma} \leq r_a \leq \frac{4L}{\gamma_i} \leq \frac{8\sqrt{2}}{\nu}, \quad r_1 = r_a + \Gamma^{-1} < \frac{16}{\nu}, \quad t_2^a = t_2^{\text{in}} + r_a, \quad t_2^1 = t_2^{\text{in}} + r_1. \quad (19)$$

For existence of the crossing, the upper candidate time is inside the fixed horizon; integrating the lower rate to that time gives at least L logarithmic gain. Strict monotonicity and the intermediate value theorem give existence and uniqueness. The transition length is at most $2/(L\nu) \leq 1/(32\nu)$, so the entire transition is also inside that already justified horizon. The lower bound on r_a ensures the second activation $\eta(\Gamma r)$ equals one before the growth threshold, since $L \geq 64$.

1.5 Closing every entry bound and the complete finite statement

At the actual steering entry t_2^1 , take $P_{1\text{in}} = A_d e^{-qr_1}/K_1$ and $P_{2\text{in}} = P_2(r_1)$. Equations (17) and (19) give the initial quotient interval in Theorem 2. Also

$$\frac{8}{15} < B_{\text{in}} = \frac{A_d e^{-qr_1}}{\sigma^2} \leq 1, \quad 0 < \delta_2 = \frac{dK_2^2}{\Gamma} \leq \frac{\log 3}{6400}, \quad 0 < \delta_1 = \frac{dK_1^2}{\Gamma} \leq \delta_2 < \frac{1}{16}. \quad (20)$$

The lower parent bound is $(4/5)(2/3)$; the upper is $A_d \leq \sigma^2$. The diffusion bounds follow from (3), (11) and $K_1 < K_2$. Equations (10), (12), (17), and (20) prove every entry/design hypothesis of the direct return theorem from the original seeds and the explicit choices (2)–(3).

Theorem 1. *For each fixed original source datum and profiles above, choose $\hat{\lambda}_1$ by (2), retain the source recursion (1), and choose any $0 < d \leq d_{\max}$ in (3). The complete modified two-layer coefficient system exists through both original growth targets, both prescribed transitions, both selected vorticity returns, and a second holding interval of physical length $1/(32\Gamma_2)$. Its times, amplitudes, original coordinates, controls, and spatial support inequalities are the exact formulas above and the inverse formulas of Theorem 2. At the second return the new vorticity amplitude and horizontal phase component are zero, the first parent vorticity is strictly positive, and*

$$\log \frac{-\Theta_2(t)}{S_2} \geq L_2 + \frac{\gamma_i}{4\Gamma_2} + \frac{\log 3}{320} \quad \text{for } t_2^b \leq t \leq t_2^b + (32\Gamma_2)^{-1}. \quad (21)$$

The inherited parent is evolved throughout; no inviscid estimate is assumed to persist under diffusion. With the full profile/localization forces below these coefficients determine compact spatial solutions from the original initial datum and zero initial velocity.

Proof. The first stage was proved by the exact map at its actual threshold. The second growth and transition, their signs and all entry bounds were proved on a common explicit physical horizon. Theorem 2 therefore applies, constructs its entire trial family before taking the least root, and proves the return and nonstationary parent hold. The second growth contributes exactly L_2 to logarithmic gain. The transition contributes at least $\gamma_i/(4\Gamma_2)$ by (18). The steering and entire hold contribute at least $\log 3/320$ by that theorem and (20). Adding these three actual integrals proves (21). All finite join controls are smooth by flatness of the original step and interior support of its pulse; the solution of the smooth ODE therefore matches to every order at those joins. The spatial identities below prove the stated compact realization with every required force. $\square$

For use in those spatial bounds, throughout both stages and their hold one may use the following proved amplitude upper bounds:

$$\begin{aligned} \sup |\Theta_1| &\leq \max\{C_A P_1^*, 2\sigma^2/K_1\}, & \sup |\Omega_1| &\leq \max\{(K_1/\nu)C_A P_1^*, 9\sigma\}, \\ \sup |\Theta_2| &\leq e^{18} P_2^*, & \sup |\Omega_2| &\leq (3K_2/\sigma)e^{18} P_2^*. \end{aligned} \quad (22)$$

For layer one, use its first-stage positive multiplier bounds and the direct second-stage bounds $B \leq 2, W < 9$. For layer two, growth never exceeds its threshold; the transition has dimensionless coefficient $a_2 K_2/(\sigma\Gamma) = a_2 b_2/\Gamma^2 \leq n_i < 2$ and $v < 2$, so its logarithmic gain is at most 4. The selected steering gain is at most $12(1 + L^{-1}) < 14$, and holding only decreases temperature. These observations prove $e^{18} P_2^*$ as well as the vorticity bound from $0 \leq v \leq 3$ on the selected steering, $v < 2$ before it, and $v = 0$ on holding.

The exact second-stage diffusion exponents before steering are $dK_2^2 r_a$ and dK_2^2/Γ ; the first parent's exponent over these same intervals is $q(r_a + 1/\Gamma)$. The steering and holding exponents retain the evolving $|\zeta_2|^2$ inside the integrals in Theorem 2; together these are the complete two-stage amplitude diffusion costs.

Finally the proved viscosity range is explicitly frequency dependent. For fixed original data and Q_2 ,

$$0 < d_{\max} \leq \frac{\nu C_L \log 3}{12800\mu^2} (\log \hat{\lambda}_1) \hat{\lambda}_1^{-2Q_2} \longrightarrow 0 \quad (\hat{\lambda}_1 \longrightarrow \infty). \quad (23)$$

To verify the limit, set $y = \log \hat{\lambda}_1$ and use $e^{2Q_2 y} \geq (2Q_2 y)^2/2$, giving $ye^{-2Q_2 y} \leq 1/(2Q_2^2 y) \rightarrow 0$. This is the dependence of the range proved here, not an assertion that every larger diffusivity fails or that a modified infinite construction is impossible. The actual finite result preserves both original thresholds and controls. A fixed-positive-diffusivity iteration would require further estimates or a changed scale design; it does not follow by iterating a range whose upper bound tends to zero.

1.6 An exact scaling restriction for this growth rule

The original datum ties the physical spatial scale to the initial amplitude: $\mu = \lceil \nu/2 \rceil$ with $\nu = \sqrt{A_0} \geq 1$. Consequently

$$\frac{\mu^2}{\nu} \geq \frac{1}{2}, \quad \hat{d} = d \frac{\mu^2}{\nu} \geq \frac{d}{2}, \quad (24)$$

where $\hat{d}$ is exactly the transformed diffusion coefficient in the original map $(X, T) = (\mu x, \nu t)$. To prove the first inequality, if $1 \leq \nu \leq 2$ then $\mu = 1$ and $\mu^2/\nu = 1/\nu \geq 1/2$. If $\nu > 2$, then $\mu \geq \nu/2$, whence $\mu^2/\nu \geq \nu/4 > 1/2$. Equality occurs at the original datum value $A_0 = 4$, $\nu = 2$, $\mu = 1$; this identifies the equality case and does not change the parameters of any constructed trajectory. The second inequality follows by multiplication by $d > 0$, using the exact diffusion operator map.

For the first modified growth rule proved here, attaining $P_1^* = S_1 e^{L_1} > S_1$ requires and is equivalent to $dK_1^2 < \Gamma_1$. This follows directly from (4): at equality the temperature amplitude is constant, and above it the amplitude strictly decreases for all growth times. Therefore every instance of this same growth rule must satisfy

$$\hat{d}\hat{\lambda}_1^2 < \sin s_*, \quad d\hat{\lambda}_1^2 < 2 \sin s_*. \quad (25)$$

The first inequality is the identical physical inequality divided by ν , since $K_1 = \mu\hat{\lambda}_1$; the second follows from (24). Thus increasing A_0 while preserving the original spatial scaling cannot make $\hat{d}$ tend to zero at a fixed positive physical d . This necessary first-growth restriction is a result about the specified modified amplitude equation and original growth control. Any route past it must change an identified part of that dynamics or scale design and verify the corresponding full spatial forces. It is not an obstruction to all forced viscous flows.

2 A direct two-layer return theorem with positive physical diffusivity

This section proves a finite controlled return for the actual two-layer coefficient system, including the evolving first parent. It uses the released Alpöge–Buckmaster steering profile from Section 3.6.3, pp. 18–22, and its identification in Section 3.7.6, pp. 31–32. The estimates below are proved directly for the modified equations; no estimate for an inviscid trajectory is transferred to a diffusive one. The spatial realization of these amplitude equations requires the explicit profile and localization forces calculated in the companion sections. This section makes no assertion about an infinite sequence.

2.1 Original parameters, entry state, and control

Keep $A_0 \geq 1$, $\mu = \lambda_0 = \lceil \sqrt{A_0}/2 \rceil$, and the paper's time-scaling parameter $\nu_{AB} = \sqrt{A_0}$. The physical frequencies are $K_i = \mu\hat{\lambda}_i$, $i = 1, 2$. The prescribed source parameters $s_*, L = L_2$ and the source profiles η, η_2 are retained. Thus $0 < s_* \leq 1/8$, η is smooth nondecreasing, zero on $(-\infty, 0]$ and one on $[1, \infty)$, and $\eta_2 \in C_c^\infty((0, 1))$ is nonnegative with $\int_0^1 \eta_2 = 1$. Put $H = \|\eta_2\|_\infty$. Physical viscosity and thermal diffusivity are both $d \geq 0$. They are distinct from ν_{AB} , μ , and the shooting parameter $m \in [0, 3]$.

Let t_1 be the actual second-layer steering-entry time. Its data are

$$\begin{aligned} \alpha(t_1) &= s_*, & \zeta_1(t_1) &= e_2, & \Omega_1(t_1) &= 0, & \Theta_1(t_1) &= -P_{1in} < 0, \\ M_2(t_1) &= I, & \zeta_2(t_1) &= e(s), & \Theta_2(t_1) &= -P_{2in} < 0, \\ \Omega_2(t_1) &= -\frac{K_2}{\sigma} P_{2in} v_{in}. \end{aligned} \quad (26)$$

Here $e(s) = (\sin s, \cos s)$, $\sigma > 0$ is the actual inherited scale, $s = L\nu_{AB}/\sigma$ is the actual source second angle, and $\Gamma = \sigma \sin s$. In particular σ, s, Γ are not reset to fixed values if the preceding diffusive stage changes them. The data $P_{1\text{in}}, P_{2\text{in}}, v_{\text{in}}$ are the actual values after growth and transition. Theorem 1 proves from both original seeds and both actual threshold crossings that these values satisfy the explicit entry margins below, for its displayed nonempty range of positive d . No coefficient comparison during steering is assumed.

For compact notation define the following exact coordinate change:

$$\begin{aligned} a &= \sin s, & h_0 &= \cos s = \sqrt{1 - a^2}, & B_{\text{in}} &= \frac{K_1 P_{1\text{in}}}{\sigma^2}, \\ C_0 &= \frac{A_0 \cos s_*}{\sigma^2}, & C_1 &= \frac{A_0 \sin s_*}{\sigma^2 a}, \\ \delta_1 &= \frac{dK_1^2}{\Gamma}, & \delta_2 &= \frac{dK_2^2}{\Gamma}, & \tau &= \Gamma(t - t_1). \end{aligned} \quad (27)$$

Every factor used as a divisor in this change is displayed and positive; physical time is recovered by $t = t_1 + \tau/\Gamma$. Set $T = 1 + L^{-1}$, $T_H = T + 1/32$, and use the exact control

$$z_m(\tau) = \begin{cases} 1 - \eta(\tau), & 0 \leq \tau \leq 1, \\ -mL\eta_2(L(\tau - 1)), & 1 \leq \tau \leq T, \\ 0, & T \leq \tau \leq T_H. \end{cases} \quad Z_m(t) = az_m(\Gamma(t - t_1)). \quad (28)$$

Flatness and interior support make these definitions smooth at both joins. This is exactly the source pulse, with source pulse amplitude $mL \sin s$, entry first component a , and entry phase length one.

2.2 Exact closed equations and their inverse map

Write $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. The two phase gradients have the exact signed determinant $\det(\zeta_1, \zeta_2) = -a$. Since $|\zeta_1| = 1$, put

$$\begin{aligned} h &= \zeta_1 \cdot \zeta_2, & D_* &= h^2 + a^2 = |\zeta_2|^2, & Y &= \sqrt{D_* - a^2 z_m^2}, \\ B &= \frac{-K_1 \Theta_1}{\sigma^2}, & W &= \frac{\Omega_1}{\sigma}, & n &= B + C_0 + C_1 h. \end{aligned}$$

The symbol D_* is a positive scalar, not the parent velocity matrix. The square root selects the positive second component of ζ_2 . The complete older/geometry system in τ is

$$\begin{aligned} h' &= W, & h(0) &= h_0, \\ W' &= B \frac{Y - h z_m}{D_*} - \delta_1 W, & W(0) &= 0, \\ B' &= -C_1 W - \delta_1 B, & B(0) &= B_{\text{in}}, \\ n' &= -\delta_1 B, & n(0) &= B_{\text{in}} + C_0 + C_1 h_0. \end{aligned} \quad (29)$$

The last equation is a proved consequence of the preceding ones, not an additional independent constraint. Define $k = n/D_*$. The newest pair is recovered from

$$\begin{aligned} v' &= z_m - kv^2, & v(0) &= v_{\text{in}}, \\ P_2 &= P_{2\text{in}} \exp\left(\int_0^\tau (kv - \delta_2 D_*) dr\right), \\ \Theta_2 &= -P_2, & \Omega_2 &= -\frac{K_2}{\sigma} P_2 v. \end{aligned} \quad (30)$$

Here P_2 is recovered only on an interval where v has been proved finite. The proof below constructs that interval before using this map.

For clarity, the full geometric inverse and the older physical state are

$$\begin{aligned} \zeta_2 &= (az_m, Y), & \zeta_1 &= \left(\frac{haz_m - aY}{D_*}, \frac{hY + a^2z_m}{D_*} \right), \\ \phi_2 &= \arcsin(az_m/\sqrt{D_*}), & \phi_1 &= \phi_2 - \arctan(a/h), & \alpha &= s_* - \phi_1, \\ M_1 &= R_\alpha, & M_2 &= R_{\alpha-s_*} \begin{pmatrix} 1 & -(h-h_0)/a \\ 0 & 1 \end{pmatrix}, & \Theta_1 &= -\sigma^2 B/K_1, & \Omega_1 &= \sigma W. \end{aligned} \quad (31)$$

The expression for M_1 uses its original first-activation datum $M_1 = I$ when $\alpha = 0$ and $p_1 = e(s_*)$; it is not an artificial reset of the first deformation at t_1 .

Proof of the exact equations and inverse. The actual parent is

$$G = -A_0 R_\alpha e_2 - K_1 P_1 \zeta_1, \quad D_{<2} = \dot{\alpha} J + \Omega_1 J \zeta_1 \otimes \zeta_1, \quad P_1 = -\Theta_1.$$

The first coefficient is $a_1 = A_0 \sin s_*/K_1$, because $\phi_1 + \alpha = s_*$. Therefore the diffusive first pair gives $\dot{A}_1 = -A_0 \sin s_* \Omega_1 - dK_1^2 A_1$ and $\dot{\Omega}_1 = -A_1 \sin \phi_1 - dK_1^2 \Omega_1$, where $A_1 = K_1 P_1$. The phase equation gives $\dot{h} = a\Omega_1$. The determinant representation is $\zeta_2 = h\zeta_1 - aJ\zeta_1$; solving its two components gives (31) and $-\sin \phi_1 = a(Y - hz_m)/D_*$. Substitution and division by the nonzero physical factors in (27) prove the first three equations in (29). Differentiating $B + C_0 + C_1 h$ proves the exact cancellation $n' = -\delta_1 B$.

The full second coefficient retains the original base term:

$$a_2 = \frac{aA_1 + A_0(a \cos s_* + h \sin s_*)}{K_2 D_*} = \frac{\sigma^2 a n}{K_2 D_*}, \quad b_2 = K_2 a z_m.$$

In particular the numerator changes when the first temperature diffuses; it has not been held fixed. The equations $\dot{\Theta}_2 = a_2 \Omega_2 - dK_2^2 D_* \Theta_2$ and $\dot{\Omega}_2 = b_2 \Theta_2 - dK_2^2 D_* \Omega_2$ give exactly (30) by differentiation. Conversely differentiating its exponential and product proves these two physical equations, so the coordinate map and its inverse preserve the complete dynamics.

For the matrices, write $S = \begin{pmatrix} 1 & -(h-h_0)/a \\ 0 & 1 \end{pmatrix}$. Then $\dot{S} = \Omega_1 (J e_2 \otimes e_2) S$ because $\dot{h} = a\Omega_1$. Differentiating $R_{\alpha-s_*} S$ proves $\dot{M}_2 = D_{<2} M_2$ and $M_2(t_1) = I$. Furthermore $S^{-T}(a, h_0) = (a, h)$, proving $M_2^{-T} e(s) = \zeta_2$ with its original oriented datum. Both matrices have determinant one. Differentiating $\zeta_{2,1} = Z_m$ in its phase equation proves the exact control

$$\dot{\alpha} = F_2^{\text{ctl}} - \frac{\dot{Z}_m}{Y}, \quad F_2^{\text{ctl}} = -\frac{\Omega_1 \zeta_{1,1} (J \zeta_1 \cdot \zeta_2)}{Y} = \frac{a \Omega_1 \zeta_{1,1}}{Y}.$$

Thus no extra angular control has been substituted for the source feedback. All divisions and inverse trigonometric branches are justified by the bounds proved next. $\square$

2.3 An explicit finite parameter range

Theorem 2. Suppose the numerical entry and design parameters in (26)–(27) satisfy

$$\begin{aligned} 0 < s &\leq \arcsin(1/8), & 0 < a &\leq \frac{1}{8}, & L &\geq 64, & 3LHa &\leq \frac{1}{4}, \\ \frac{1}{2} &\leq B_{\text{in}} \leq 2, & 0 < C_0 &\leq \frac{1}{4}, & 0 < C_1 &\leq \frac{1}{32}, \\ \frac{1}{2} &\leq v_{\text{in}} \leq 2, & 0 &\leq \delta_1 \leq \frac{1}{16}. \end{aligned} \quad (32)$$

These are explicit inequalities on actual entry data and fixed design numbers, not hypotheses on a subsequently unknown trajectory. The displayed range of s fixes the original positive-cosine branch; the inequality on $\sin s$ alone would not fix that branch. Then the entire family $m \in [0, 3]$

exists through $\tau = T$, with $P_2 > 0$, and its endpoint map $m \mapsto v(T; m)$ is continuous. It satisfies the quantitative endpoint margins

$$v(T; 0) \geq \frac{1}{7}, \quad v(T; 3) \leq -\frac{1}{480}. \quad (33)$$

The zero set is a nonempty compact subset of $(0, 3)$ and hence has a least element m_* . Every zero satisfies

$$\frac{13}{96} \leq m_* \leq 3 - \frac{1}{480}, \quad W(T; m_*) \geq \frac{13h_0}{3840} \exp(-\delta_1/L) > 0. \quad (34)$$

No monotonicity of the endpoint map or uniqueness of the root is claimed.

The selected state continues through the nonzero holding interval $[T, T_H]$, with $\zeta_{2,1} = \Omega_2 = 0$ and with the first parent still evolving according to (29). On the whole selected interval $[0, T_H]$,

$$\begin{aligned} h_0 \leq h < 3, \quad 0 \leq W < 9, \quad \frac{191}{512} < B \leq 2, \quad \frac{1}{4} < n < 3, \\ \frac{63}{64} \leq D_* < 10, \quad Y \geq \frac{\sqrt{59}}{8}, \quad \zeta_{1,2} > \frac{1}{16}, \quad \frac{1}{40} < k < 4. \end{aligned} \quad (35)$$

The same geometry bounds hold throughout every trial through T . Every trial has $|v| < 6$ through T , so its linear amplitude pair never loses the negative sign of Θ_2 .

The exact selected logarithmic gain at every $\tau_f \in [T, T_H]$ is

$$\log \frac{P_2(\tau_f)}{P_{2\text{in}}} = \int_0^T kv \, d\tau - \delta_2 \int_0^{\tau_f} D_* \, d\tau, \quad \frac{\log 3}{160} - 10\delta_2\tau_f \leq \log \frac{P_2(\tau_f)}{P_{2\text{in}}} \leq 12T. \quad (36)$$

In particular the strictly positive physical range

$$0 < d \leq \min \left\{ \frac{\Gamma}{16K_1^2}, \frac{\Gamma \log 3}{6400K_2^2} \right\} \quad (37)$$

gives $\log(P_2(\tau_f)/P_{2\text{in}}) \geq \log 3/320 > 0$. The interval is expressed in actual physical units. Theorem 1 supplies an explicit positive $d_{\max}$ using only the original data and chosen frequencies, and proves (32) and (37) for every $0 < d \leq d_{\max}$ while retaining the dependence of the actual entry data and Γ on d .

Proof. First construct geometry without using the newest amplitude pair. On an initial interval where $B > 0$, the bracket $Y - hz_m$ is nonnegative. If $0 \leq z_m \leq 1$, this follows by squaring: $Y^2 - h^2 z_m^2 = D_*(1 - z_m^2) \geq 0$; if $z_m \leq 0$, it follows from $Y, h > 0$. Consequently the integrating-factor formula for W gives $W \geq 0$. Therefore $h \geq h_0$ and $B' = -C_1 W - \delta_1 B \leq 0$, giving $B \leq B_{\text{in}}$. The control bound is $|az_m| \leq 1/4$, so

$$Y^2 \geq h_0^2 - \frac{1}{16} \geq \frac{59}{64} > 0.$$

For $0 \leq \tau \leq 1$, $Y - hz_m \leq \sqrt{D_*}$, hence $W' \leq B_{\text{in}}/h_0$ and $h \leq h_0 + B_{\text{in}}\tau^2/(2h_0)$. On the negative pulse the same argument gives

$$W' \leq \frac{B_{\text{in}}}{h_0} \{1 + mL\eta_2(L(\tau - 1))\}.$$

During holding its right side is B_{in}/h_0 . Integration on any interval up to T_H therefore yields

$$W \leq \frac{B_{\text{in}}}{h_0} (T_H + 3) \leq \frac{26}{3} < 9,$$

because $h_0 \geq 15/16$, $T_H \leq 17/16$, and $B_{\text{in}} \leq 2$. The duration after the first part is at most $L^{-1} + 1/32 \leq 3/64$, so

$$h \leq h_0 + \frac{B_{\text{in}}}{2h_0} + \frac{B_{\text{in}}}{h_0} (T_H + 3) \frac{3}{64} \leq 1 + \frac{16}{15} + \frac{13}{32} = \frac{1187}{480} < 3.$$

No quotient variable has entered these estimates.

The exact integrating-factor identity for the first temperature is

$$B(\tau) = e^{-\delta_1 \tau} B_{\text{in}} - C_1 \int_0^\tau e^{-\delta_1(\tau-r)} W(r) dr.$$

Since $W \geq 0$, $h' = W$, $\delta_1 T_H \leq 17/256$, and $e^{-x} \geq 1 - x$ for $x \geq 0$, it follows that

$$B(\tau) \geq \frac{1}{2} \left(1 - \frac{17}{256} \right) - \frac{1}{32} (h(\tau) - h_0) > \frac{239}{512} - \frac{3}{32} = \frac{191}{512} > \frac{1}{4}.$$

This strict estimate excludes a first exit through $B = 0$. All square-root denominators stay uniformly positive and h, W, B remain bounded. The smooth finite-dimensional vector field is therefore bounded with bounded first derivatives on a compact neighborhood of every such solution. Local existence, uniqueness and the integral equation extend the solution through T_H ; the same neighborhood can be chosen uniformly for $m \in [0, 3]$. Differentiating that integral equation and applying the elementary Gronwall estimate proves continuous dependence on m on this interval. This constructs the entire older/geometry trial family before the newest pair.

Now $n = B + C_0 + C_1 h > 1/4$, and $n' = -\delta_1 B \leq 0$, so $n \leq B_{\text{in}} + C_0 + C_1 h_0 \leq 73/32 < 3$. The asserted D_* and k bounds follow. For the first phase, $|az_m| \leq 1/4$, $a \leq 1/8$, $h \geq 15/16$ and $Y > 7/8$ give

$$\zeta_{1,2} = \frac{hY + a^2 z_m}{D_*} > \frac{(15/16)(7/8) - 1/32}{10} = \frac{101}{1280} > \frac{1}{16}.$$

Thus the branches and control denominators in (31) stay in their specified domains. This proves all geometry claims.

Construct the Riccati variable next. On $[0, 1]$, its forcing $z_m = 1 - \eta$ is nonnegative. The reciprocal differential inequality $(1/v)' \leq 4$, initially while $v > 0$, gives

$$\frac{v_{\text{in}}}{1 + 4v_{\text{in}}\tau} \leq v(\tau) \leq v_{\text{in}} + \tau \leq 3. \quad (38)$$

The positive lower bound rules out a first zero, and the upper bound follows by integrating $v' \leq 1$. These bounds also exclude finite escape. Put $v_d = v(1)$. In particular $v_d \geq 1/6$. Moreover

$$\int_0^1 kv^2 d\tau \geq \frac{1}{40} \int_0^1 \frac{v_{\text{in}}^2}{(1 + 4v_{\text{in}}\tau)^2} d\tau = \frac{v_{\text{in}}^2}{40(1 + 4v_{\text{in}})} \geq \frac{1}{480}.$$

The last inequality follows because $x^2/(1 + 4x)$ is increasing for $x > 0$, as its derivative is $(2x + 4x^2)/(1 + 4x)^2 > 0$. Integrating the Riccati equation proves

$$\frac{1}{6} \leq v_d \leq 3 - \frac{1}{480}. \quad (39)$$

On the pulse put $V(y) = v(1 + y/L)$ and $M_2(y) = \int_0^y \eta_2(r) dr$. Its exact integral equation is

$$V(y) = v_d - m M_2(y) - \frac{1}{L} \int_0^y k(1 + r/L) V(r)^2 dr.$$

Under the temporary bound $|V| \leq 6$, it gives $V \leq v_d < 3$ and $V \geq -3 - 144/L \geq -21/4 > -6$. The bound is strict initially and strictly improves itself at every possible first exit, so $|V| < 6$ on the full pulse. The quotient has therefore been constructed with a uniform finite bound, and the exponential formula (30) constructs a negative Θ_2 solving the entire original linear pair. Its uniqueness follows by subtracting two solutions of that linear pair and applying the integral Gronwall inequality to their zero initial difference. In particular the construction agrees with the unique linear-pair solution and has not assumed absence of a temperature zero.

For $m = 0$, the pulse solves $v' = -kv^2$, so $v(T; 0) \geq v_d/(1 + 4v_d/L) \geq 1/(6 + 4/L) > 1/7$. For $m = 3$, the integral equation and (39) give $v(T; 3) \leq v_d - 3 \leq -1/480$. The uniformly

finite quotient and geometry imply continuous dependence of $v(T; m)$ on m by their integral equations. The intermediate value theorem proves existence of a zero. Its zero set is closed in $[0, 3]$ and excludes both endpoints, proving the stated compactness and least-root selection.

At any selected zero, $v' = -mL\eta_2 - kv^2 \leq 0$ throughout the pulse. Thus $0 \leq v \leq v_d$ there, and

$$m = v_d - \int_1^T kv^2 d\tau \in \left[v_d - \frac{4v_d^2}{L}, v_d \right] \subseteq \left[\frac{13}{96}, 3 - \frac{1}{480} \right].$$

For the lower inclusion use $v_d \leq 3$, $L \geq 64$ and $v_d \geq 1/6$. On the pulse,

$$W' + \delta_1 W = B \frac{Y + hmL\eta_2}{D_*} \geq \frac{h_0}{40} mL\eta_2 (L(\tau - 1)).$$

Its integrating-factor formula, $W(1) \geq 0$ and the unit integral of η_2 prove (34). On holding, $z_m = 0$ and $v(T) = 0$, so uniqueness gives $v = 0$. The older geometry continues by the already established bounds, and (30) gives the exact remaining temperature decay.

Finally, (38) gives

$$\int_0^1 kv d\tau \geq \frac{1}{40} \int_0^1 \frac{v_{\text{in}}}{1 + 4v_{\text{in}}\tau} d\tau = \frac{1}{160} \log(1 + 4v_{\text{in}}) \geq \frac{\log 3}{160}.$$

The selected pulse adds nonnegative gain, the hold adds zero gain, and $v \leq 3$, $k < 4$ bound the gain above by $12T$. The exact diffusion exponent is at most $10\delta_2\tau_f$. This proves (36); substitution of (37) gives its positive lower bound. $\square$

2.4 Returned state, holding dynamics, and diffusion costs

At the selected physical return time $t_b = t_1 + T/\Gamma$ one has

$$\zeta_2(t_b) = (0, \sqrt{D_*(T)}), \quad \Omega_2(t_b) = 0, \quad \zeta_1(t_b) = \frac{(-a, h(T))}{\sqrt{D_*(T)}}, \quad \alpha(t_b) = s_* + \arctan(a/h(T)),$$

and $\Theta_1(t_b) = -\sigma^2 B(T)/K_1 < 0$, $\Omega_1(t_b) = \sigma W(T) > 0$. The first parent is therefore neither frozen nor returned to zero. Its full motion creates the specified returned state of the second wave. The controlled holding interval has physical duration $1/(32\Gamma)$. On it the complete system is

$$h' = W, \quad W' = B/\sqrt{h^2 + a^2} - \delta_1 W, \quad B' = -C_1 W - \delta_1 B, \\ P_2(\tau) = P_2(T) \exp\left(-\delta_2 \int_T^\tau (h(r)^2 + a^2) dr\right).$$

These formulas retain the evolving phase length inside the diffusion integral. They also prove the exact stationarity of $\zeta_{2,1}, \Omega_2$ throughout this finite hold.

The returned parent vorticity also gives positive scalar curl of the full local velocity at the origin. Indeed $z_m = z'_m = 0$ on holding, so the proved control and inverse give the exact physical formulas

$$\dot{\alpha} = -\frac{a^2 \Omega_1}{h^2 + a^2}, \quad \omega(0, t) = 2\dot{\alpha} + \Omega_1 = \Omega_1 \left(1 - \frac{2a^2}{h^2 + a^2}\right) \geq \frac{31}{32} \Omega_1 > 0. \quad (40)$$

For the inequality use $h \geq h_0$, $h_0^2 + a^2 = 1$ and $a^2 \leq 1/64$. The positive forcing in the holding equation for W gives $W(\tau) \geq W(T)e^{-\delta_1(\tau-T)} > 0$ by its integrating factor. These statements include the return endpoint because the original pulse is identically zero near that endpoint. The compact spatial construction has constant envelopes and exact linear profiles near the origin, so its curl there is precisely the quantity in (40); no cutoff derivative enters it.

The total first-wave damping exponents on the ramp, pulse and hold are δ_1 , δ_1/L and $\delta_1/32$, respectively. The corresponding second-wave exponents are exactly

$$\delta_2 \int_0^1 D_* d\tau, \quad \delta_2 \int_1^T D_* d\tau, \quad \delta_2 \int_T^{T_H} D_* d\tau.$$

Each exponent lies between $(63/64)\delta_2$ and $10\delta_2$ times its displayed interval length. In original physical coordinates these equal $dK_2^2 \int |\zeta_2(t)|^2 dt$, with $K_2 = \mu \hat{\lambda}_2$ and $dt = d\tau/\Gamma$; none of the factors μ , ν_{AB} , K_i or Γ has been dropped. The exact first-amplitude identity in the proof also records the additional parent-response term $-C_1 \int e^{-\delta_1(\tau-r)} W(r) dr$; a pure common-damping multiplier would miss that term.

For the parent affine fields the altered gradient is not independently prescribed. With $B_1 = -\sigma^2 B \zeta_1$, direct differentiation gives

$$\dot{G} + D_{<2}^T G = -dK_1^2 B_1, \quad C = 2\dot{\alpha} + \Omega_1, \quad \dot{C} - G_1 = 2\ddot{\alpha} - A_0 \sin \alpha - dK_1^2 \Omega_1.$$

Thus $G \cdot x$, $D_{<2} x$ have scalar force $(-dK_1^2 B_1) \cdot x$ and full vector force $\frac{1}{2}(2\ddot{\alpha} - A_0 \sin \alpha - dK_1^2 \Omega_1) Jx$ with pressure $-\frac{1}{2}x^T \text{sym}(\dot{D}_{<2} + D_{<2}^2 - e_2 \otimes G)x$. Their Laplacians vanish. These affine formulas are local exact fields; the compact original-profile realization and its additional spatial forces are supplied by the accompanying finite-profile construction. They must not be discarded when interpreting the amplitude result as a statement about the full spatial equations.

Here is a direct verification of the affine force and pressure, including their signs. Put $D = D_{<2}$, $u(x, t) = D(t)x$ and $\theta(x, t) = G(t) \cdot x$. Both summands in D have zero trace, so $\text{div } u = 0$ and $D^2 = -\det(D)I$ by multiplication of a general trace-free 2×2 matrix. The scalar residual is exactly $(\dot{G} + D^T G) \cdot x$. In that derivative the term $K_1 \dot{\Theta}_1 \zeta_1$ cancels $(\Omega_1 J \zeta_1 \otimes \zeta_1)^T (-A_0 R_\alpha e_2)$, apart from $-dK_1^2 B_1$; the angular terms cancel because $\dot{\zeta}_1 = \dot{\alpha} J \zeta_1$ and $\partial_t (R_\alpha e_2) = \dot{\alpha} J R_\alpha e_2$. Also $\zeta_1 \cdot J \zeta_1 = 0$ eliminates the self-interaction. For momentum let $\mathcal{A} = \dot{D} + D^2 - e_2 \otimes G$. Then $\partial_t u + (u \cdot \nabla)u - \theta e_2 = \mathcal{A}x$, whereas the displayed pressure contributes $-\text{sym}(\mathcal{A})x$. Their sum is $\text{skew}(\mathcal{A})x$. The scalar curl of Dx is $C = D_{21} - D_{12}$; since D^2 is scalar and the curl of $(G \cdot x)e_2$ is G_1 ,

$$\text{skew}(\mathcal{A}) = \frac{\dot{C} - G_1}{2} J.$$

Finally $\dot{\Omega}_1 = -K_1 P_1 \zeta_{1,1} - dK_1^2 \Omega_1$ and $G_1 = A_0 \sin \alpha - K_1 P_1 \zeta_{1,1}$ give precisely the force stated above. Both affine Laplacians are zero by direct second differentiation. Thus these are the full vector equations with the specified pressure, not merely a vorticity residual.